

MATH 112A - Mathematical Biology

Dani Nguyen

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1 Review

1.1 Ordinary Differential Equations

An ODE can be written as $F(y, y', y'', \dots, y^{(k)}) = 0$. The order of the ODE is the order of the highest derivative.

Linear ODEs are a special class of ODEs that have the form:

$$a_0(t)y + a_1(t)y' + \dots + a_k(t)y^{(k)} = f(t)$$

There exists two special cases:

1. If $f(t) = 0$ then the linear ODE is called homogeneous.
2. If $a_i(t)$ are constant and $f(t) = 0$ then we have a linear homogeneous ODE with constant coefficients.

ex

$$y' = 0$$

This is a first order linear homogeneous ODE with constant coefficients.

$$y''' + \sin t \cdot y'' + (t^2 + 1)y + 20t = 0$$

This is a third order linear non-homogeneous ODE

$$y' \cdot y + 20$$

This is a first order non-linear non-homogeneous ODE.

$$y^2 + y'' = 0$$

This is a second order non-linear homogeneous ODE.

The simplest type of ODEs is linear, homogeneous, with constant coefficients.

$$\boxed{ex} \quad y'' + y' - 6y = 0$$

Look for solutions as:

$$\begin{aligned} y &= e^{pt} \\ y' &= pe^{pt} \\ y'' &= p^2 e^{pt} \end{aligned}$$

$$p^2 e^{pt} + pe^{pt} - 6e^{pt} = 0$$

$$p^2 + p - 6 = 0$$

$$\begin{aligned} p &= \frac{-1 \pm 5}{2} \\ &= -3 \text{ or } 2 \end{aligned}$$

The general solution is $y(t) = Ae^{-3t} + Be^{2t}$ Initial value problem: Suppose

$$y(0) = 1$$

$$y'(0) = 0$$

Then use the general solution to find A and B

$$y' = -3Ae^{-3t} + 2Be^{2t}$$

$$\begin{cases} y(0) = Ae^{-3 \cdot 0} + Be^{2 \cdot 0} = A + B = 1 \\ y'(0) = -3A + 2B = 0 \end{cases}$$

Solving the linear system gives $A = \frac{2}{5}$ and $B = \frac{3}{5}$

$$\boxed{ex} \quad y' + 2y - e^t = 0$$

$$p(t) = 2 \quad q(t) = e^t$$

$$E(t) = \exp \int_0^t p(t) dt = e^{2t}$$

$$\begin{aligned}
e^{2t}y' + 2e^{2t}y &= e^{2t} \cdot e^t \\
\frac{d}{dx}e^{2t}y &= e^{3t} \\
e^{2t}y &= \frac{1}{3}e^{3t} + c \\
y &= \frac{1}{3}e^t + ce^{-2t}
\end{aligned}$$

Separable ODEs

$$\begin{aligned}
\frac{dy}{dt} &= g(y)h(t) \\
\frac{dy}{g(y)} &= h(t)dt \\
\int \frac{dy}{g(y)} &= \int h(t)dt
\end{aligned}$$

A more realistic model of population growth where r = growth rate and k = carrying capacity.

$$\frac{dy}{dt} = ry\left(1 - \frac{y}{k}\right)$$

When $y \ll k$, $\text{RHS} \approx ry$